19CS549 - UI & UX Design

a) Explore the fundamentals of UI & UX

Introduction:-

In the digital world, users interact with applications, web sites, and software daily. To ensure these interactions are meaningful, easy and efficient, user interface (UI) & User experience (UX) design play a crucial role. UI & UX together focus on creating products that are not only visually appealing but also functional and user-friendly

Fundamentals of user Interface (UI):

Definition of UI:-

User interface (UI) refers to the visual elements & interactive components through which a user communicates with a digital product.

Core UI elements:

i) Layout:-

Arrangement of elements on the screen

ii) Color:-

Creates visual hierarchy & represents brand identity

iii) Typography:-

Font type, size, spacing

iv) Buttons & icons

Enable interaction

v. Consistency

Uniform design across screens

vi) Responsiveness

Works across devices

Fundamentals of UX :-

Definition of UX:-

UX refers to the overall feeling and satisfaction a user gets with a product, from start to finish

Key UX components :-

i) User research:-

Understanding user needs using surveys, interviews, personas

ii) Information architecture:

Organizing content logically & navigation structure

iii) Usability:-

Ease of use

iv) Accessibility:-

Designing for all users

v) User flow:

Step by step journey of the user

vi) Interaction design:

How users interact with elements

b) Differentiate between UI & UX using real-world case studies

Introduction:-

UI & UX are two closely related but distinct concepts in digital product design. While UI focuses on the visual appearance & interactive elements, UX focuses on the overall experience & satisfaction of the user. The difference becomes clear when observed through real-world case-studies

Basic difference difference b/w UI & UX

| Aspect | UI | UX |
|---|---|---|
| Meaning | Visual & interactive design | Overall user journey |
| Focus | Look & feel | Usability & satisfaction |
| Role | Makes product attractive | Makes product meaningful |

Case study :

i) Google search engine

UX perspective

- User can search with minimal effort
- Fast response time

UI perspective

- Simple search bar
- Clean white background

ii) Amazon Shopping app

UX perspective

- Easy product discovery
- Filters & sorting options

UI perspective

- Product cards with images
- "Add to cart" & "Buy now" buttons

iii) ATM machine

UX perspective

- Step by Step transaction flow
- Clear instructions

UI perspective

- Outdated screens
- Physical buttons

iv) Food delivery apps

UX perspective

- Easy restaurant search
- Simple ordering process

UI perspective

- Bright brand colors
- Clear call to active buttons